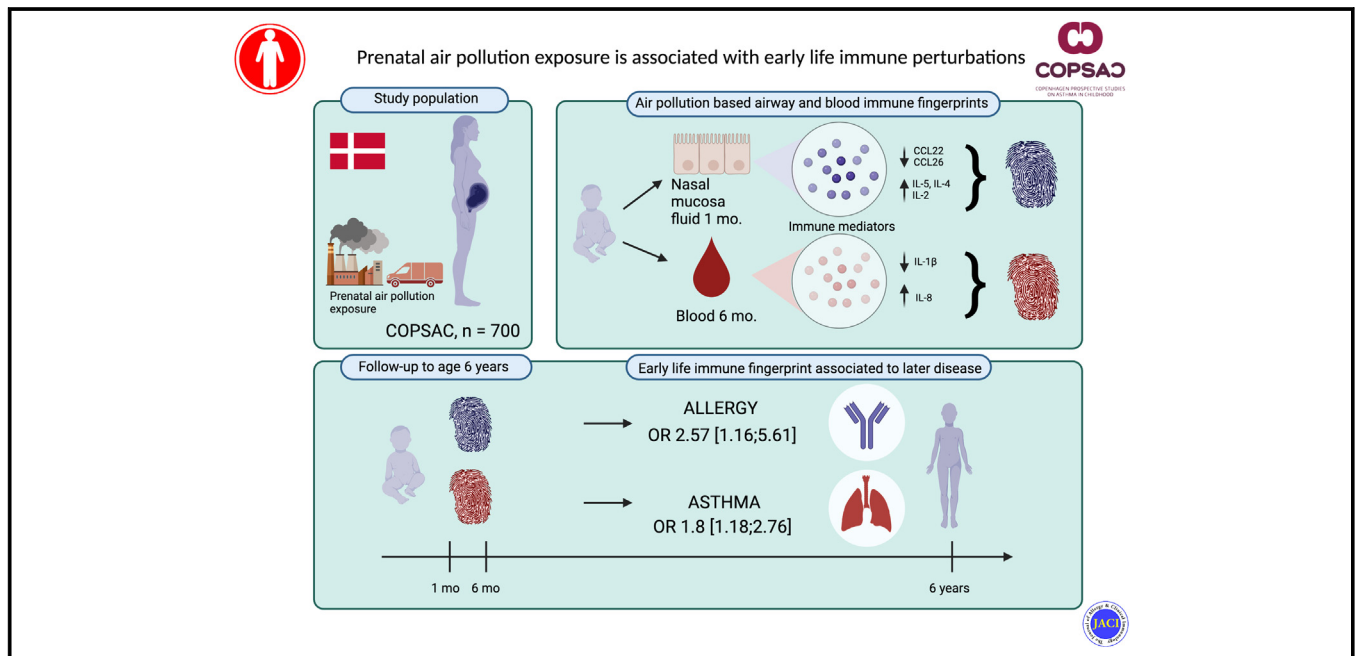


Prenatal exposure to ambient air pollution is associated with early life immune perturbations



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GRAPHICAL ABSTRACT



Background: Exposure to ambient air pollution has been linked to asthma, allergic rhinitis, and other inflammatory disorders, but little is known about the underlying mechanisms. **Objective:** We studied the potential mechanisms leading from prenatal ambient air pollution exposure to asthma and allergy in childhood.

Methods: Long-term exposure to nitrogen dioxide (NO₂) as well as to particulate matter with a diameter of ≤ 2.5 and ≤ 10 μm (PM_{2.5} and PM₁₀) were modeled at the residence level from conception to 6 years of age in 700 Danish children followed clinically for development of asthma and allergy. Nasal mucosal immune mediators were assessed at age 4 weeks and 6 years,

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inflammatory markers in blood at 6 months, and nasal epithelial DNA methylation and gene expression at age 6 years.

Results: Higher prenatal air pollution exposure with NO₂, PM_{2.5}, and PM₁₀ was associated with an altered nasal mucosal immune profile at 4 weeks, conferring an increased odds ratio [95% confidence interval] of 2.68 [1.58, 4.62] for allergic sensitization and 2.63 [1.18, 5.81] for allergic rhinitis at age 6 years, and with an altered immune profile in blood at age 6 months conferring increased risk of asthma at age 6 years (1.80 [1.18, 2.76]). Prenatal exposure to ambient air pollution was not robustly associated with immune mediator, epithelial DNA methylation, or gene expression changes in nasal cells at age 6 years.

Conclusion: Prenatal exposure to ambient air pollution was associated with early life immune perturbations conferring risk of allergic rhinitis and asthma. These findings suggest potential mechanisms of prenatal exposure to ambient air pollution on the developing immune system. (J Allergy Clin Immunol 2023;151:212-21.)

Key words: Air pollution, asthma, allergy, childhood, environment, epigenetic, inflammation, immune, sensitization, traffic, urbanization

Long-term exposure to ambient air pollution in early life has been suggested to have detrimental effects on respiratory health in children^{1,2} and has been associated with an increased risk of childhood asthma in Denmark.³ Evidence of associations between nitrogen dioxide (NO₂), as well as particulate matter with diameters of ≤ 2.5 μm (PM_{2.5}) and ≤ 10 μm (PM₁₀), and respiratory diseases in children⁴⁻⁷ are growing, but the underlying mechanisms are not fully understood, particularly in relation to prenatal and early life exposure and early life mechanisms.

Both short-term and long-term air pollution exposure can induce oxidative stress and potentially affect the immune system and result in a more proinflammatory response across immune cell types.⁸⁻¹² Long-term exposure to ambient air pollution has also been associated with changes in the epigenetic profiles of placenta cells and blood cells from school-age children.¹³⁻¹⁷

Previous studies have shown that the nasal epithelial epigenome may serve as a sensitive biomarker of asthma, allergy, and airway disease.^{18,19} Thus, changes in DNA methylation (DNAm) leading to dysregulation of gene expression in these cells is a potential mechanism mediating long-term health effects from early life exposure.

Here, we investigated the effects of pre- and postnatal exposure to ambient air pollution on the early life nasal mucosal immune profiles and systemic inflammation, and changes in the epigenome of nasal epithelial cells and nasal mucosal immune profiles at 6 years of age as potential mechanisms of air pollution exposure in relation to asthma and allergy development. This was done in the Danish COPEnhagen Prospective Studies on Asthma in Childhood (COPSAC) 2010 mother-child cohort, where children were followed prospectively for asthma and related traits from birth to 6 years of age.

METHODS

Study population

The COPSAC₂₀₁₀ cohort is an ongoing prospective mother-child cohort comprising 700 children born to women residing in Zealand, Denmark, with

Abbreviations used

CI:	Confidence interval
COPSAC:	Copenhagen Prospective Studies on Asthma in Childhood
DMC:	Differently methylated CpGs
DNAm:	DNA methylation
FDR:	False discovery rate
IQR:	Interquartile range
OR:	Odds ratio
PC:	Principal component
PM _{2.5/10} :	Particulate matter with an aerodynamic diameter of $\leq 2.5/\leq 10$ μm
sPLS-DA:	Sparse partial least squares discriminant analysis

enrollment in week 24 of pregnancy. This cohort has been followed intensely with several scheduled and acute visits to the research unit, where the pediatricians were solely responsible for the diagnosis and treatment of respiratory, allergic, and skin-related symptoms.²⁰⁻²² The study included randomized interventions with fish oil and high-dose vitamin D during pregnancy. Fig E1 (in the Online Repository available at www.jacionline.org) provides an overview of the ages of sampling for the included end points, and Fig E2 (also in the Online Repository) shows a flowchart of the included analyses and corresponding sample sizes.

Governance

We are aware of and comply with recognized codes of good research practice, including the Danish Code of Conduct for Research Integrity. We comply with national and international rules on the safety and rights of patients and healthy subjects, including Good Clinical Practice as defined in the EU's Directive on Good Clinical Practice, the International Conference on Harmonisation's good clinical practice guidelines, and the Helsinki Declaration. Privacy is important to us, which is why we follow national and international legislation on General Data Protection Regulation, the Danish Act on Processing of Personal Data, and the practice of the Danish Data Inspectorate.

Assessment of ambient air pollution

We used the unique personal identification numbers assigned by the Danish civil registration system to link the COPSAC data with individual-level maternal and child home address history from conception until the latest day of follow-up, including the dates of moving, emigration, immigration, and death.²³

Concentrations of ambient PM_{2.5}, PM₁₀, and NO₂ were modeled at each home address using the Danish Eulerian Hemispheric Model (DEHM)—urban background model (UBM)—Danish Air Pollution and Human Exposure Modelling System (AirGIS) model system.²⁴⁻²⁶ The local contribution from traffic at the street level was calculated by the Operational Street Pollution Model.²⁷ The local contribution from sources of primary anthropogenic emitted particles was calculated with a 1 km $\times$ 1 km resolution using the UBM.²⁴ Finally, the contribution from the long-range transport and regional background due to nonlocal sources were calculated using the DEHM with a 5.6 km $\times$ 5.6 km resolution.²⁵ For each participant, we calculated time-weighted mean from conception to birth (prenatal exposure), from birth to the DNAm sampling point (total postnatal exposure), and in the year before the age 6-year sampling point (prior year exposure). In addition, we calculated 3 short-term exposure means: the first 4 weeks of life (postnatal first 4 weeks), the first 6 months of life (postnatal first 6 months) and the month before the age 6-year sampling point (last month) for covariates in our models. No more than 10% missing values were allowed for each of the time-weighted means. We fitted models with increments for each exposure, separately, corresponding to the interquartile range (IQR) for the full study population (see Table E1 in the Online Repository at www.jacionline.org). Because of the high

correlation between air pollutants, we also calculated a combined air pollution variable using the first principal component (PC1) for the 3 pollutants in each of the exposure windows (explaining >83% of the variance).

Immune mediators, DNAm, and gene expression

Nasal mucosal lining fluid samples were collected at age 4 weeks and 6 years using a synthetic absorptive matrix to assess airway immune mediators.^{28,29}

Inflammatory markers in blood was assessed at 6 months of age using high-sensitivity immunoassays based on electrochemiluminescence in a multiplex setting³⁰ DNA methylation (Illumina Infinium Methylation EPIC array; Illumina, San Diego, Calif) and gene expression (RNA sequencing) were assessed in respiratory (nasal) epithelium at age 6 years.¹⁹

An overview of samples (Fig E2) and methodologic details can be found in the Methods in the Online Repository available at www.jacionline.org.

Allergy traits and asthma

Allergic sensitization at age 6 years was defined as either a positive skin prick test result of at least 3 mm (ALK-Abello) or a serum level of specific IgE against common inhalant allergens of at least 0.35 kUa/L (ImmunoCAP; Thermo Fisher Scientific, Waltham, Mass).

Allergic rhinitis at 6 years of age was based on allergic sensitization and interview of the parents regarding the history of troublesome nasal congestion, sneezing, or a runny nose outside periods with common colds and in the relevant period of the sensitized aeroallergen.

Asthma ever from birth up to 6 years of age was diagnosed prospectively through a detailed quantitative symptom algorithm.^{31,32} Current asthma was defined as an active diagnosis (need of controller medication) in the 6th year of life.

Statistical analysis

Immune mediators were normalized according to the total sum per sample to account for high collinearity between immune mediators within samples. The resulting values were log transformed and *z* scored. We multiple-test adjusted for the number of immune mediators in the single-component analysis, as we had already documented a strong correlation among air pollution components.

In order to provide a combined immune “fingerprint” reflecting the immune changes associated with air pollution exposure, we implemented a supervised multivariate analysis in sparse partial least squares discriminant analysis (sPLS-DA) models.³³ This was used to explore associations between the early life airway (4 weeks) and systemic (6 months) immune mediators and air pollution measurements. We used the PC1 of air pollution for these analyses. Loading plots of the included immune mediators can be seen in Figs E3 and E4 in the Online Repository at www.jacionline.org.

To assess the differently methylated CpGs (DMC) between individuals, we used an adjusted linear model within the limma framework. Enrichment analysis was performed by a hypergeometric test and assessed using a false discovery rate (FDR) of 5%. We similarly used the limma framework to analyze differentially expressed genes.³⁴ DMC and differentially expressed genes were reported using both a FDR of 5% and a more liberal FDR of 15%.

Associations between exposure to air pollution, allergy, and asthma end points were assessed using multivariate logistic regression models adjusted for sex, maternal smoking during pregnancy (yes or no), maternal exposure to passive smoking during pregnancy (0, 1, 2, or 3 trimesters), passive smoking during early childhood (more than 1 year, yes or no), parity (number of birthed children), social attributes of the mother (age at birth, income and education level), and mode of delivery (normal or cesarean section). Analyses of prenatal air pollution exposure were further adjusted for postnatal air pollution exposure.

Because urban/rural living is associated with air pollution exposure as well as early life inflammation, asthma, and allergy,³⁵ we performed subanalyses adjusting for a continuous measure of living in an urban area based on land use data (urbanicity).

Additional details and covariate descriptions are available in the Online Repository available at www.jacionline.org.

Ethics

This study was approved by the local ethics committee (H-B-2008-093) and was registered at the Danish Data Protection Agency (2015-41-3696). Parents provided informed consent for their children’s participation before enrollment.

RESULTS

Study characteristics

The prenatal mean (IQR) concentrations of ambient NO₂, PM_{2.5}, and PM₁₀ were 18.8 (8.3), 10.1 (1.5), and 15.3 (2.6) µg/m³, respectively (Table E1). There were highly positive correlations between the pollutants, particularly between prenatal exposure to PM_{2.5} and PM₁₀ (Pearson *r* = 0.96, *P* < .001). The correlations between urbanicity at birth and prenatal exposure to PM_{2.5}, PM₁₀, and NO₂ was low to modest, with Pearson *r* = 0.29, 0.22, and 0.66, respectively (see Fig E5 in the Online Repository at www.jacionline.org).

The study population characteristics of the 700 COPSAC₂₀₁₀ children and their mothers as well as for the subset of individuals with DNAm data are summarized in Table E2 (available in the Online Repository at www.jacionline.org). At 6 years of age, 146 children (22.4%) had been diagnosed with asthma at any time, 117 (23.7%) were sensitized to inhalant allergens, 48 (7.4%) currently had asthma, and 44 (6.7%) currently had allergic rhinitis (Table E2).

Early life nasal mucosal immune profile

We associated prenatal air pollution exposure with the nasal mucosal immune mediators in mucosal fluid at 4 weeks and observed significant associations with several cytokines after multiple test correction, primarily for cytokines contributing to type 2 response. Higher exposure to each of the 3 air pollution components was significantly associated with lower levels of CCL22 and CCL26. Furthermore, higher prenatal NO₂ exposure was associated with increased levels of IL-4 and IL-5, PM₁₀ exposure was nominally associated with increased IL-5 and IL-2, and PM_{2.5} exposure was nominally associated with increased IL-13. Generally, the 3 air pollutants showed similar directions of association (ie, reduced levels of CCL22 and CCL26, and increased levels of IL-5, IL-4, and IL-2) (Fig 1; see Table E3 in the Online Repository at www.jacionline.org). Adjusting for urbanicity generally resulted in unchanged associations, except for stronger and now nominally significant associations between exposure to PM₁₀ and increased IL-5 and IL-2 (see Fig E6 in the Online Repository).

To further explore the effects of prenatal air pollution on the age 4-week nasal mucosal immune profile and how this was related to disease development, we used supervised multivariate analysis, sPLS-DA, to generate an early life nasal mucosal immune fingerprint of prenatal exposure to air pollution. This fingerprint was associated with increased risk of allergic sensitization at 6 years (odds ratio [95% confidence interval], or OR [95% CI]) (2.68 [1.58, 4.62], *P* < .001), and allergic rhinitis at 6 years (2.63 [1.18, 5.81], *P* = .017), while there was a trend of association with current asthma at 6 years (2.01 [0.93, 4.19], *P* = .067) (Table 1). Adjusting for urbanicity and postnatal air

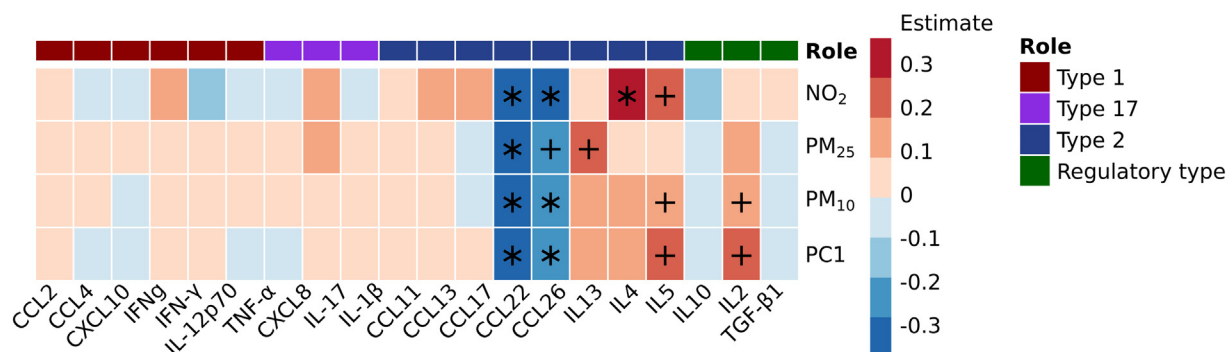


FIG 1. Association of prenatal exposure to ambient air pollution with nasal mucosal immune mediators at age 4 weeks. Estimates are expressed as standard deviation changes from the mean of immune mediator levels per IQR changes of prenatal exposure to ambient air pollution. Models included the following covariates: birth season, vitamin D intervention, an atopy diagnosis for the mother, older children, infection with *Haemophilus influenzae*, *Moraxella catarrhalis*, *Staphylococcus aureus*, or *Streptococcus pneumoniae*, and social attributes of the mother (age of mother at birth, income and education level) as well as postnatal air pollution exposure (postnatal first 4 weeks). *Bonferroni significance ($P < .003$). +Nominal significance ($P < .05$). The most prominent immune type of involvement of the cytokines is listed at right. Table E3 provides effect estimates and confidence intervals.

TABLE I. Prenatal ambient air pollution fingerprint on nasal mucosa immune mediators at 4 weeks of age in relation to asthma and allergy phenotypes at age 6 years

Characteristic	4-week airway immunity (sPLS) adjusted for:					
	Covariates			Covariates including postnatal air pollution		
	OR [95% CI]	P value	N (n)	OR [95% CI]	P value	N (n)
Sensitization to aerosol inhalants at 6 years	2.61 [1.56, 4.44]	<.001*	330 (67)	2.68 [1.58, 4.62]	<.001*	330 (67)
Allergic rhinitis 6-year cross-sectional	2.57 [1.16, 5.61]	.017*	286 (23)	2.63 [1.18, 5.81]	.016*	286 (23)
Current asthma at 6 years	2.36 [1.15, 4.73]	.016*	406 (27)	2.01 [0.93, 4.19]	.067	406 (27)
Asthma 0-6 years of age	1.48 [0.95, 2.30]	.083	406 (88)	1.35 [0.85, 2.13]	.19	406 (88)

sPLS, Sparse partial least squares model (immune mediator fingerprint), refers to a multivariate classification based on a prenatal air pollution exposure as the outcome and nasal mucosal immune mediators sampled at age 4 weeks as the predictors. Covariates included in these models were: smoking during pregnancy, passive smoking, passive smoking during early childhood, parity, social attributes of the mother (age of mother at birth, income and education level), receipt of cesarian section, and sex. Postnatal air pollution represents exposure in first 6 years of life (total postnatal exposure) and is included as a covariate.

*Nominally statistically significant.

pollution exposure (total postnatal exposure) did not change the observed associations substantially (see Table E4 in the Online Repository at www.jacionline.org).

Early life systemic inflammation

Prenatal air pollution exposure was strongly associated with systemic inflammatory markers measured in blood at age 6 months (Fig 2), with increased levels of CXCL8 (IL-8) and decreased levels of IL-1 β across all air pollution components, while increased NO₂ exposure was associated with decreased levels of IL-6. These associations were largely unchanged after adjusting for urbanicity (see Fig E7 in the Online Repository at www.jacionline.org).

As for the nasal mucosal immune mediators, we generated an early life systemic immune fingerprint of prenatal air pollution (PC1). This systemic fingerprint was significantly associated with increased asthma risk at age 6 years (OR [95% CI] 1.80 [1.18, 2.76], $P = .006$), but not with allergic rhinitis or allergic sensitization (Table 2). Adjusting for postnatal air pollution exposure (total postnatal exposure) attenuated the association with asthma (1.54 [0.99, 2.40], $P = .056$) (Table 2), while adjusting for urbanicity did not change the associations substantially (see Table E5 in the Online Repository at www.jacionline.org).

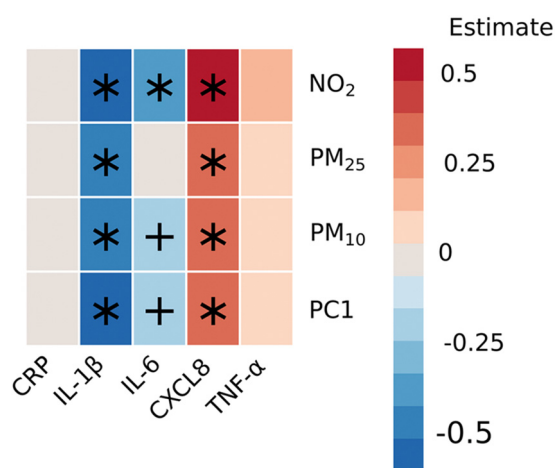


FIG 2. Association of prenatal exposure to ambient air pollution with inflammatory markers in blood at age 6 months. Estimates are expressed as standard deviation changes from the mean of immune marker levels per IQR increase of prenatal exposure to ambient air pollution. Models included the following covariates: birth season and postnatal air pollution exposure (postnatal first 6 months). Table E5 provides the effect estimates and confidence intervals. *Bonferroni significance ($P < .01$). +Nominal significance ($P < .05$).

TABLE II. Prenatal ambient air pollution fingerprint on systemic inflammatory markers at 6 months in relation to asthma and allergy phenotypes at age 6 years

Characteristic	Systemic immune (sPLS) adjusted for:					
	Covariates			Covariates including postnatal air pollution		
	OR [95% CI]	P value	N (n)	OR [95% CI]	P value	N (n)
Sensitization to aerosol inhalants at 6 years	1.18 [0.72, 1.93]	.50	389 (82)	1.18 [0.72, 1.94]	.50	389 (82)
Allergic rhinitis 6-year cross-sectional	1.08 [0.47, 2.40]	.85	331 (25)	1.05 [0.45, 2.37]	.90	331 (25)
Current asthma at 6 years	1.36 [0.67, 2.71]	.39	471 (31)	1.19 [0.57, 2.41]	.64	471 (31)
Asthma 0–6 years of age	1.80 [1.18, 2.76]	.006*	471 (105)	1.70 [1.11, 2.62]	.015*	471 (105)

sPLS, Sparse partial least squares model (immune mediator fingerprint), refers to a multivariate classification based on prenatal air pollution exposure as the outcome and blood immune mediators sampled at age 6 months as the predictors. Covariates included in these models were: smoking during pregnancy, passive smoking exposure during pregnancy, passive smoking during early childhood, parity, social attributes of the mother (age of mother at birth, income and education level), cesarian section, and sex. Postnatal air pollution represents exposure in the first 6 years of life (total postnatal exposure) and is included as a covariate.

*Nominally statistically significant.

Nasal mucosal immune profile at 6 years

There were no significant associations between prenatal air pollution exposure and nasal mucosal immune mediators at 6 years of age (see Fig E8 in the Online Repository at www.jacionline.org).

There were also no significant associations between postnatal air pollution exposure (total postnatal exposure) and immune mediator levels at 6 years of age after adjustment for multiple testing (see Fig E9 in the Online Repository at www.jacionline.org), and this was also the case when restricting the exposure window to the year before sampling at age 6 years (prior year exposure) (see Fig E10 in the Online Repository).

Nasal epithelial gene expression at 6 years

We identified associations between prenatal air pollution exposure and gene expression in nasal epithelial cells at 6 years of age. Specifically, higher prenatal NO₂ exposure led to increased expression of *AKAP9* (LogFC 0.23) after FDR 5% adjustment, Fig 3, A and B). When using an FDR of 15%, PM_{2.5} was associated with lower expression of *LINC00644* (LogFC −0.66) and *RP11-420K14* (LogFC −0.82) and higher expression of *CCL2* (LogFC 0.72) (Fig 3, C and D), while increased prenatal PM₁₀ exposure was associated with lower expression of *RP11-420K14* (LogFC −0.77) (Fig 3, E and F). There were no statistically significant associations between the 4-week nasal mucosal immune mediator fingerprint (data not shown), or postnatal air pollution exposure (total postnatal exposure) and gene expression (see Fig E11 in the Online Repository at www.jacionline.org).

DNAm in nasal epithelial cell at 6 years

We found 1 differentially methylated CpG site in relation to prenatal air pollution exposure using a 5% FDR threshold, specifically in relation to PM_{2.5} levels. This site was also identified for prenatal air pollution exposure to PM₁₀ using a 15% FDR (see, in the Online Repository at www.jacionline.org, Figs E12, E13, and E14, as well as Table E6). We observed 2 DMC for postnatal air pollution exposure (total postnatal exposure) using the 15% FDR, while we saw 4 and 10 DMC for postnatal air pollution exposure restricted to the year before sampling prior year exposure. One site was found across all 3 air pollution exposure windows: cg11603096, which is located upstream in the transcription start site of the gene *EIF3D* (Table E6). We did not

observe associations between gene expression levels for the genes closest to the identified CpG sites and air pollution exposure (Table E6). We did not see any association between our age 4-week nasal mucosal immune mediator fingerprint and DNAm levels looking genome-wide (data not shown).

Enrichment analyses using sets of previously identified DMC associated with asthma and allergy showed no significant enrichment regardless of the exposure window (see Table E7 in the Online Repository at www.jacionline.org). Similarly, we found no significant enrichment in sets of DMC previously associated with air pollution in blood or placental cells (see Table E8 in the Online Repository).

Asthma and allergy end points

There were no significant direct associations between prenatal exposure to ambient air pollution and asthma and allergy end points at age 6 years, although the direction of association was of increased OR with increased exposure across all disease end points except for allergic rhinitis and PMs (Table 3). After adjustment for postnatal air pollution exposure, we observed a nominally significant association between NO₂ and allergic sensitization (OR [95% CI] 1.76 [1.06, 2.94], *P* = .029) (Table 3).

Higher postnatal exposure (0–6 years) to ambient air pollution was associated with an increased risk of having current asthma by age 6 for all 3 air pollution components (Table 4). However, except for the association between PM₁₀ exposure and current asthma, all associations were nonsignificant after adjusting for urbanicity (see Table E9 in the Online Repository at www.jacionline.org).

Higher air pollution exposures in the 6th year of life (prior year exposure) were associated with increased risk of current asthma by age 6 (eg, for NO₂ exposure: OR [95% CI] 1.70 [1.22, 2.34], *P* = .001) (Table 4). The prior year exposure in Table 4 refers to 4.6 µg/m³ increments, which are smaller than 5.2 µg/m³ used for 0–6-year average total postnatal exposure. These associations remained after adjusting for urbanicity (see Table E10 in the Online Repository at www.jacionline.org).

DISCUSSION

Main findings

We found strong associations between prenatal air pollution exposures and both the nasal mucosal immune profile and the blood inflammatory profile shortly after birth. The profiles

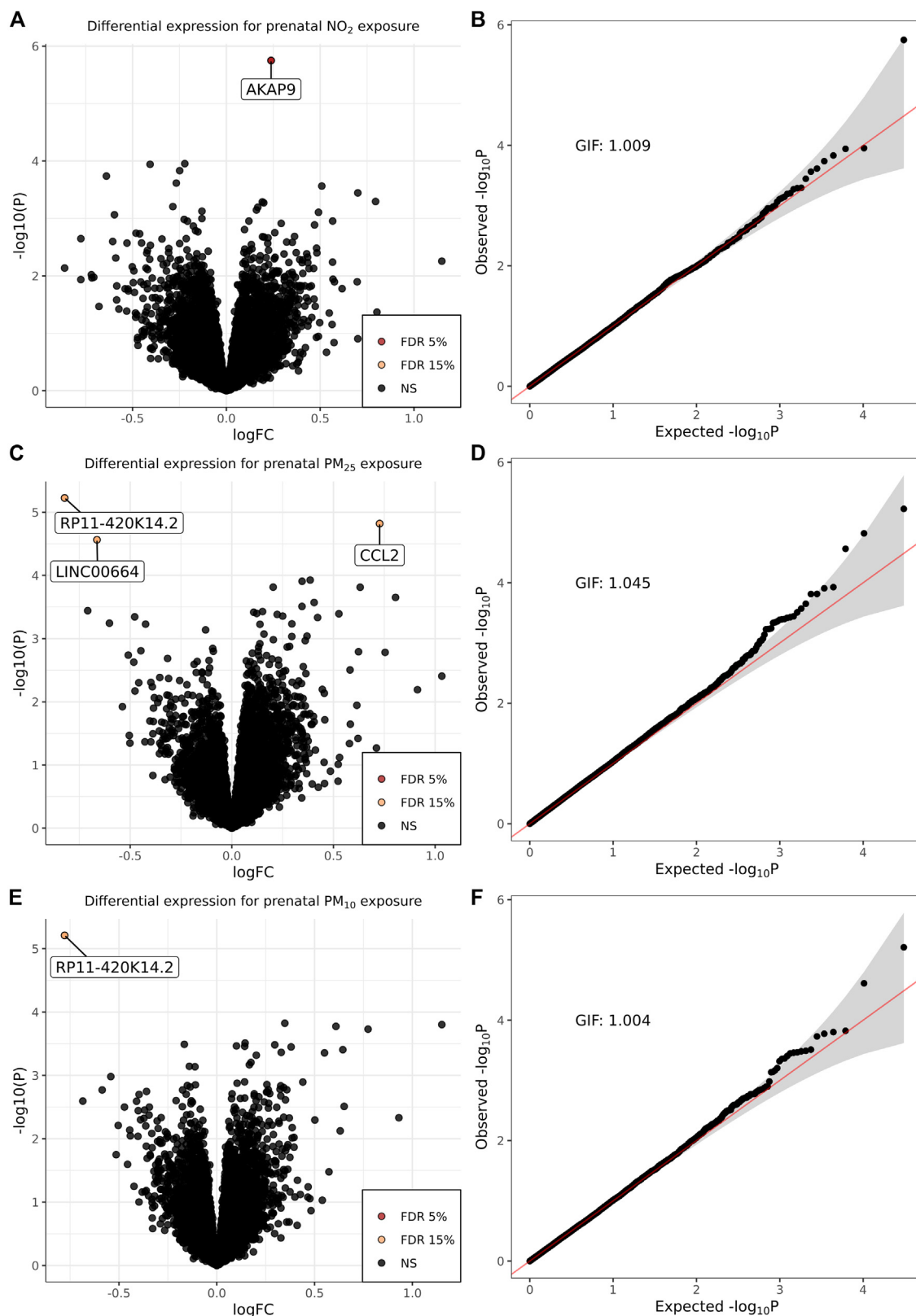


FIG 3. Gene expression in nasal epithelial cells at age 6 years and prenatal air pollution exposure. Volcano plot of differential gene expression in children exposed to prenatal air pollution exposure ($n = 381$). Models included the following covariates: RNA concentration, RNA integrity number, site of sampling, and season of sampling. *Left*, Volcano plots of the 15,363 genes included in this study. *Right*, QQplots showing theoretical and observed distributions of P values. The confidence intervals of the uniform distribution in the QQplots are calculated using a beta distribution, and the genomic inflation factor (λ) is included in the plot. (A and B) NO₂ exposure. (C and D) PM_{2.5} exposure. (E and F) PM₁₀ exposure.

TABLE III. Prenatal ambient air pollution exposure in relation to asthma and allergy phenotypes at age 6 years

Characteristic	NO2		PM2.5		PM10		N (n)
	OR [95% CI]	P value	OR [95% CI]	P value	OR [95% CI]	P value	
Prenatal							
Sensitization to aerosol inhalants at 6 years	1.27 [0.91; 1.77]	.15	1.11 [0.77; 1.62]	.57	1.09 [0.76; 1.55]	.65	412 (87)
Allergic rhinitis 6-year cross-sectional	1.12 [0.62; 1.92]	.69	0.92 [0.49; 1.72]	.80	0.85 [0.45; 1.56]	.60	353 (28)
Current asthma at 6 years	1.34 [0.82; 2.11]	.22	1.33 [0.75; 2.34]	.32	1.35 [0.79; 2.25]	.26	503 (33)
Asthma 0-6 years of age	1.22 [0.91; 1.63]	.17	1.09 [0.78; 1.50]	.62	1.11 [0.81; 1.51]	.52	503 (113)
Prenatal air pollution exposure adjusted for postnatal air pollution exposure							
Sensitization to aerosol inhalants at 6 years	1.76 [1.06; 2.94]	.029*	1.16 [0.77; 1.76]	.48	1.16 [0.75; 1.78]	.50	412 (87)
Allergic rhinitis 6-year cross-sectional	1.18 [0.48; 2.70]	.70	0.89 [0.44; 1.75]	.73	0.80 [0.38; 1.63]	.55	353 (28)
Current asthma at 6 years	0.74 [0.35; 1.46]	.40	0.92 [0.49; 1.74]	.80	0.88 [0.46; 1.64]	.68	503 (33)
Asthma 0-6 years of age	0.96 [0.61; 1.48]	.85	0.91 [0.62; 1.32]	.61	0.89 [0.60; 1.30]	.54	503 (113)

Covariates included in these models were: smoking during pregnancy, passive smoking exposure during pregnancy, passive smoking during early childhood, parity, social attributes of the mother (age of mother at birth, income and education level), cesarean section, and sex. Postnatal air pollution represents exposure in the first 6 years of life (total postnatal exposure) and is included as a covariate.

*Nominally statistically significant.

TABLE IV. Postnatal ambient air pollution exposure in relation to asthma and allergy phenotypes at age 6 years

Characteristic	NO ₂		PM _{2.5}		PM ₁₀		N (n)
	OR [95% CI]	P value	OR [95% CI]	P value	OR [95% CI]	P value	
Postnatal air pollution exposure							
Sensitization to aerosol inhalants at 6 years	1.11 [0.84; 1.46]	.45	0.9 [0.69; 1.16]	.45	0.94 [0.72; 1.20]	.65	404 (85)
Allergic rhinitis 6-year cross-sectional	1.17 [0.73; 1.78]	.47	0.84 [0.51; 1.28]	.44	1.04 [0.67; 1.50]	.85	346 (27)
Current asthma at 6 years	1.73 [1.19; 2.50]	.003*	1.51 [1.08; 2.07]	.012*	1.56 [1.14; 2.09]	.004*	495 (33)
Asthma 0-6 years of age	1.39 [1.09; 1.77]	.008*	1.24 [1.00; 1.53]	.048*	1.23 [1.00; 1.51]	.043*	495 (111)
Prior year air pollution exposure							
Sensitization to aerosol inhalants at 6 years	1.1 [0.85; 1.40]	.45	0.97 [0.72; 1.28]	.82	1.02 [0.81; 1.25]	.88	346 (86)
Allergic rhinitis 6-year cross-sectional	1.12 [0.71; 1.64]	.61	0.86 [0.51; 1.41]	.56	1.04 [0.69; 1.47]	.84	496 (27)
Current asthma at 6 years	1.7 [1.22; 2.34]	.001*	1.68 [1.17; 2.40]	.004*	1.54 [1.17; 2.00]	.001*	405 (33)

Covariates included in these models were: smoking during pregnancy, passive smoking, passive smoking during early childhood, parity, social attributes of the mother (age of mother at birth, income and education level), cesarean section, and sex.

*Nominally statistically significant.

associated with higher prenatal air pollution exposure were associated with allergic sensitization, allergic rhinitis, and asthma determined by age 6 years, independent of postnatal air pollution exposure. There was little evidence of association between pre- or postnatal air pollution exposure and immune mediators, DNAm, or gene expression at age 6 years. These findings point to an independent effect from prenatal air pollution exposure on the developing upper airway and systemic immune system in early life with potential long-term effects on the risk of developing allergy and asthma.

Strengths and limitations

The main strength of this study was the combined assessments of early life upper airway immune profiles, systemic inflammation, gene expression, and DNAm in the children with prospective clinical follow-up of asthma and allergy development.²² The infants were continuously monitored for the development of asthma and allergy symptoms at the research unit according to predefined diagnostic and treatment procedures, increasing homogeneity in diagnoses.

Another strength of our study was the assessment of ambient air pollution exposure using comprehensive high-resolution modeling of ambient air pollution at residential addresses from conception to end of follow-up using detailed high-quality data. With this, we were able to consider changes of addresses and the time lived at each address, enabling the evaluation of both pre- and postnatal air pollution exposure.

It is novel and a strength that we measured upper airway immune mediators, and DNAm and gene expression in the nasal epithelial cells, which might be a more relevant target tissue than blood when studying air pollution effects in relation to asthma and allergy.³⁶

Finally, it is an advantage that we have detailed information on potential confounders, including detailed data on smoking, vitamin D intake, and land cover-based urbanicity. Factors related to urban/rural living are important potential confounders in studies of air pollution exposure because these are strongly correlated. This is particularly important in studies of asthma and allergy where urban living is a suspected risk factor,³⁵ potentially through altered microbial exposures, as previously reported from the COPSAC₂₀₁₀ cohort.³⁵

There are some limitations to our study. The sample size of this study was limited compared to the sample size of thousands of individuals normally required to show associations between air pollution and clinical outcomes. However, the sample size was like that of other studies reporting associations between ambient air pollution and mechanistic data.^{15,17} We therefore hypothesized that we would still have power to show association with mechanistic data, including immune data, and thereby help understanding the mechanistic effects of air pollution exposure. Furthermore, the more detailed phenotyping might outweigh the limited sample size, and indeed we were able to detect some direct associations between air pollution exposure and clinical end points. Because of the many genes and methylation sites tested, we cannot exclude the notion that gene expression and methylation changes mediating air pollution effects are present but could not be detected as a result of a lack of statistical power.

Interpretation

We report strong associations between prenatal air pollution exposure and upper airway immune profiles at 4 weeks of age. Changes were particularly seen for the atopy-related type 2 immune mediators, with increased levels of IL-13, IL-4, and IL-5 and reduced levels of CCL22 and CCL26. There was overlap in the direction of associations for the 3 different air pollutants (PM_{2.5}, PM₁₀, NO₂) for several immune mediators, suggesting similarities in effects, but also some differences—for example, in relation to IL-4 and IL-5, where we only observed higher levels for NO₂ exposure. In Denmark, variation in NO₂ levels is mainly driven by traffic emissions, whereas long-range transport contributes significantly to total PM_{2.5}, with the highest levels in the southern parts of the country.^{25,26}

Using a multivariate approach, we further showed that the prenatal air pollution exposure–derived fingerprint of nasal mucosal immune mediators in early life was associated with an increased risk of sensitization to inhalant aeroallergens and allergic rhinitis at 6 years of age. This suggests that the changes in upper airway immune mediators after prenatal air pollution exposure might have detrimental and long-lasting effects on the risk of developing allergic sensitization and allergic disease in the airways. This might be the underlying mechanism of previously reported associations between air pollution exposure at birth and allergic sensitization to inhalant allergens in later childhood.^{37,38}

In line with the current findings, we recently reported associations between reduced levels of CCL26 in the airways at 4 weeks and increased risks of allergic sensitization and allergic rhinitis at age 6 years in the COPSAC₂₀₁₀ cohort.³⁹ CCL26 is reported as a chemoattractant for CCR3-expressing immune cells typically involved in type 2 airway disease, such as eosinophils and basophils.³⁹ Our findings of lower CCL26 in early life associated with later allergic airway disease emphasizes the complex, and still poorly understood, immune mechanisms underlying allergy development.

Prenatal air pollution exposure was also strongly correlated with systemic inflammatory markers at 6 months of age, which remained robust after adjustment for both postnatal air pollution exposure and urbanicity. The associations were in the direction of increased levels of CXCL8 (IL-8) and decreased levels of IL-1β across higher levels of all 3 air pollution components, and increased levels of TNF-α with increasing exposure to PM_{2.5} and PM₁₀. Previous studies have shown associations between

prenatal PM_{2.5} and PM₁₀ exposure and C-reactive protein⁴⁰ and IL-6 levels in blood early childhood.⁴¹ We did not observe associations with C-reactive protein, and we report negative associations between NO₂ exposure and IL-6, but we also note that the previously reported associations were for children at age 4 years.⁴¹

The prenatal air pollution exposure–derived fingerprint of inflammatory blood markers in early life was associated with an increased risk of asthma by age 6 years; the signal persisted after adjustment for postnatal exposure. This suggests that the systemic inflammatory changes after prenatal air pollution exposure might have detrimental and long-lasting effects on the risk of asthma. We speculate that these effects on the developing immune system could also be involved in previously reported associations between prenatal exposure to air pollution and chronic inflammatory disorders in adulthood, such as chronic obstructive pulmonary disease^{42,43} and cardiovascular disease.^{44–46}

The only association between air pollution exposure and nasal gene expression that survived correction for multiple testing (FDR < 0.05) was between prenatal exposure to NO₂ and *AKAP9*. Another suggestive association (FDR < 0.15) was between prenatal exposure to PM_{2.5} and *CCL2*. Both of these genes have been associated with inflammation in mouse models,⁴⁷ while elevated *CCL2* expression levels have also been shown in asthmatic individuals.⁴⁸ These findings are in line with the long-term programming effects of prenatal exposure to air pollution² but need to be replicated in independent studies.

We observed limited evidence of association between air pollution exposure and differential DNAm in airway epithelium at age 6 years, given that there is little literature support for the relevance of the top associated CpGs and little consistency between associations seen for the individual pollutants. We did observe a single demethylated CpG site located in the transcriptional start site of the eukaryotic translation initiation factor-3 gene (*EIF3D*), possibly increasing the expression of this gene, although we could not see this in our gene expression data. *EIF3D* has been reported to be involved in the general activation or repression of translation in mammalian cells by means of messenger RNA binding,⁴⁹ and it likely serves an important role in T-cell differentiation,⁵⁰ so it thus may impair an adequate inflammatory response. The lack of robust methylation signals suggests that potential epigenetic signals after ambient air pollutant exposures are subtle and/or of a temporal nature or restricted to cells from other tissues, such as blood or lower airway cells. Thus, further studies might benefit from studying these tissues. This is further supported by the lack of replication of air pollution related CpGs previously identified in a similar study using DNAm data from children with a mean age of 12.9.³⁶

The associations between air pollution and clinical end points in the current study fit with the direction of effect for recurrent wheeze and asthma in early infancy reported in a previous Danish study.⁵ In particular, exposure to air pollution in the first 6 years of life and the year before the 6-year visit was associated with a diagnosis of asthma. Adjusting for urbanicity attenuated the results for 0–6 years of exposure, while results for prior year exposure were largely unchanged. The effect sizes observed were higher than reported in previous studies,^{3,5} possibly as a result of our finer resolution in air pollution modeling, more comprehensive adjustment for potential confounders, and/or more uniform asthma definition available in our COPSAC₂₀₁₀ cohort. A meta-analysis of 5 similarly sized European birth cohorts relying on

land-use regression modeled air pollution and parent-reported doctor-diagnosed asthma in children aged 4 and 8 did not observe such associations.¹²

We adjusted for urban/rural living using a continuous urbanicity gradient. Although this adjustment attenuated associations between postnatal air pollution exposure and asthma, the remaining associations between ambient air pollution exposure and the early life upper airway immune profile, systemic inflammatory markers, asthma, and allergy were largely unaffected. This indicates that the associations between prenatal air pollution exposure and early life immunology and later airway disease were not due to confounding from urban versus rural living.

Conclusion

We report an association between prenatal exposure to ambient air pollution in terms of NO₂, PM_{2.5}, and PM₁₀ and the early life airway immune profile, which was related to later development of allergic airway disease. Prenatal exposure to air pollution was also associated with early life systemic inflammation, and the associated systemic profile was associated with later asthma. These findings suggest potential mechanisms of prenatal exposure to ambient air pollution on the developing immune system increasing the risk of childhood allergy and asthma and emphasize that air pollution exposure during pregnancy can have independent, long-term health effects in offspring.

We express our deepest gratitude to the children and families of the COPSAC₂₀₁₀ cohort study for all their support and commitment. We acknowledge and appreciate the unique efforts of the COPSAC research team members.

Clinical implications: There is a detrimental effect of air pollution exposure during pregnancy on the developing immune system of offspring, with long-term effects on the risk of allergic rhinitis and asthma.

REFERENCES

- Saha P, Johny E, Dangi A, Shinde S, Brake S, Eapen M, et al. Impact of maternal air pollution exposure on children's lung health: an Indian perspective. *Toxics* 2018;6:68. <https://doi.org/10.3390/toxics6040068>.
- Johnson NM, Hoffmann AR, Behlen JC, Lau C, Pendleton D, Harvey N, et al. Air pollution and children's health—a review of adverse effects associated with prenatal exposure from fine to ultrafine particulate matter. *Environ Health Prev Med* 2021;26:72. <https://doi.org/10.1186/s12199-021-00995-5>.
- Pedersen M, Liu S, Andersen ZJ, Andersen AMN, Bisgaard H, Brandt J, et al. Long-term exposure to ambient air pollution and asthma incidence. Available at: <https://www.healtheffects.org/research/ongoing-research/impact-exposure-air-pollution-asthma-multi-exposure-assessment#pedersen>. Accessed September 19, 2022.
- Bonato M, Gallo E, Bazzan E, Marson G, Zagolin L, Cosio MG, et al. Air pollution relates to airway pathology in children with wheezing. *Ann Am Thorac Soc* 2021;18:2033-40.
- Holst GJ, Pedersen CB, Thygesen M, Brandt J, Geels C, Bønnelykke JH, et al. Air pollution and family related determinants of asthma onset and persistent wheezing in children: nationwide case-control study. *BMJ* 2020;370:m2791.
- Cohen AJ, Ross Anderson H, Ostro B, Pandey KD, Krzyzanowski M, Künzli N, et al. The global burden of disease due to outdoor air pollution. *J Toxicol Environ Health A* 2005;68:1301-7.
- Mazur M, Czarnobilska M, Dya W, Czarnobilska E. Trends in the epidemiology of allergic diseases of the airways in children growing up in an urban agglomeration. *J Clin Med* 2022;11:2188.
- Glencross DA, Ho TR, Camiña N, Hawrylowicz CM, Pfeffer PE. Air pollution and its effects on the immune system. *Free Radic Biol Med* 2020;151:56-68. <https://doi.org/10.1016/j.freeradbiomed.2020.01.179>.
- Gruzdeva O, Merid SK, Gref A, Gajulapuri A, Lemonnier N, Ballereau S, et al. Exposure to traffic-related air pollution and serum inflammatory cytokines in children. *Environ Health Perspect* 2017;125:067007.
- Klümper C, Krämer U, Lehmann I, von Berg A, Berdel D, Herberth G, et al. Air pollution and cytokine responsiveness in asthmatic and non-asthmatic children. *Environ Res* 2015;138:381-90.
- Gogna P, King WD, Villeneuve PJ, Kumarathasan P, Johnson M, Lanphear B, et al. Ambient air pollution and inflammatory effects in a Canadian pregnancy cohort. *Environ Epidemiol* 2021;5:e168. <https://doi.org/10.1097/ee9.00000000000000168>.
- Fuertes E, Sunyer J, Gehring U, Porta D, Forastiere F, Cesaroni G, et al. Associations between air pollution and pediatric eczema, rhinoconjunctivitis and asthma: a meta-analysis of European birth cohorts. *Environ Int* 2020;136:105474.
- Plusquin M, Guida F, Polidoro S, Vermeulen R, Raaschou-Nielsen O, Campanella G, et al. DNA methylation and exposure to ambient air pollution in two prospective cohorts. *Environ Int* 2017;108:127-36. <https://doi.org/10.1016/j.envint.2017.08.006>.
- Neven KY, Saenen ND, Tarantini L, Janssen BG, Lefebvre W, Vanpoucke C, et al. Placental promoter methylation of DNA repair genes and prenatal exposure to particulate air pollution: an ENVIRON AGE cohort study. *Lancet Planet Health* 2018;2:e174-83. [https://doi.org/10.1016/s2542-5196\(18\)30049-4](https://doi.org/10.1016/s2542-5196(18)30049-4).
- Gruzdeva O, Xu CJ, Breton CV, Annesi-Maesano I, Antó JM, Auffray C, et al. Epigenome-wide meta-analysis of methylation in children related to prenatal NO₂ air pollution exposure. *Environ Health Perspect* 2017;125:104-10.
- Ji N, Fang M, Baptista A, Cepeda C, Greenberg M, Mincey JC, et al. Exposure to traffic-related air pollution and changes in exhaled nitric oxide and DNA methylation in arginase and nitric oxide synthase in children with asthma. *Environ Health* 2021;20:12.
- Prunicki M, Cauwenberghs N, Lee J, Zhou X, Movassagh H, Noth E, et al. Air pollution exposure is linked with methylation of immunoregulatory genes, altered immune cell profiles, and increased blood pressure in children. *Sci Rep* 2021;11:4067.
- Cardenas A, Sordillo JE, Rifas-Shiman SL, Chung W, Liang L, Coull BA, et al. The nasal methylome as a biomarker of asthma and airway inflammation in children. *Nat Commun* 2019;10:3095.
- Morin A, McKennan CG, Pedersen CET, Stokholm J, Chawes BL, Malby Schoos AM, et al. Epigenetic landscape links upper airway microbiota in infancy with allergic rhinitis at 6 years of age. *J Allergy Clin Immunol* 2020;146:1358-66.
- Bisgaard H, Stokholm J, Chawes BL, Vissing NH, Bjarnadóttir E, Schoos AMM, et al. Fish oil-derived fatty acids in pregnancy and wheeze and asthma in offspring. *N Engl J Med* 2016;375:2530-9.
- Chawes BL, Bønnelykke K, Stokholm J, Vissing NH, Bjarnadóttir E, Schoos AMM, et al. Effect of vitamin D₃ supplementation during pregnancy on risk of persistent wheeze in the offspring: a randomized clinical trial. *JAMA* 2016;315:353-61.
- Bisgaard H, Vissing NH, Carson CG, Bischoff AL, Følsgaard NV, Kreiner-Møller E, et al. Deep phenotyping of the unselected COPSAC₂₀₁₀ birth cohort study. *Clin Exp Allergy* 2013;43:1384-94. <https://doi.org/10.1111/cea.12213>.
- Pedersen CB. The Danish civil registration system. *Scand J Public Health* 2011;39(7 suppl):22-25.
- Brandt J, Christensen JH, Frohn LM, Palmgren F, Berkowicz R, Zlatev Z. Operational air pollution forecasts from European to local scale. *Atmos Environ* 2001;35:S91-8.
- Brandt J, Silver JD, Frohn LM, Geels C, Gross A, Hansen AB, et al. An integrated model study for Europe and North America using the Danish Eulerian Hemispheric Model with focus on intercontinental transport of air pollution. *Atmos Environ* 2012;53:156-76.
- Khan J, Kakosimos K, Raaschou-Nielsen O, Brandt J, Jensen SS, Ellermann T, et al. Development and performance evaluation of new AirGIS—a GIS based air pollution and human exposure modelling system. *Atmos Environ* 2019;198:102-21.
- Jensen SS, Kettel M, Becker T, Christensen J, Brandt J, Plejdurp M, et al. High resolution multi-scale air quality modelling for all streets in Denmark. *Transp Res Part D Trans Environ* 2017;52:322-39.
- Chawes BLK, Edwards MJ, Shamji B, Walker C, Nicholson GC, Tan AJ, et al. A novel method for assessing unchallenged levels of mediators in nasal epithelial lining fluid. *J Allergy Clin Immunol* 2010;125:1387-9.e3.
- Wolsk HM, Chawes BL, Thorsen J, Stokholm J, Bønnelykke K, Brix S, et al. Noninvasive sampling of mucosal lining fluid for the quantification of *in vivo* upper airway immune-mediator levels. *J Vis Exp* 2017;126:55800. <https://doi.org/10.3797/55800>.
- Chawes BLK, Stokholm J, Bønnelykke K, Brix S, Bisgaard H. Neonates with reduced neonatal lung function have systemic low-grade inflammation. *J Allergy Clin Immunol* 2015;135:1450-6.e1.

31. Bisgaard H, Pedersen S, Anhøj J, Agertoft L, Hedlin G, Gulsvik A, et al. Determinants of lung function and airway hyperresponsiveness in asthmatic children. *Respir Med* 2007;101:1477-82.
32. Bisgaard H, Pipper CB, Bønnelykke K. Endotyping early childhood asthma by quantitative symptom assessment. *J Allergy Clin Immunol* 2011;127:1155-64.e2.
33. Rohart F, Gautier B, Singh A, Lê Cao KA. mixOmics: an R package for 'omics feature selection and multiple data integration. *PLoS Comput Biol* 2017;13:e1005752.
34. Ritchie ME, Phipson B, Wu D, Hu Y, Law CW, Shi W, et al. limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res* 2015;43:e47.
35. Lehtimäki J, Thorsen J, Rasmussen MA, Hjelmsø M, Shah S, Mortensen MS, et al. Urbanized microbiota in infants, immune constitution, and later risk of atopic diseases. *J Allergy Clin Immunol* 2021;148:234-43.
36. Sordillo JE, Cardenas A, Qi C, Rifas-Shiman SL, Coull B, Luttmann-Gibson H, et al. Residential PM_{2.5} exposure and the nasal methylome in children. *Environ Int* 2021;153:106505.
37. Sordillo JE, Webb T, Kwan D, Kamel J, Hoffman E, Milton DK, et al. Allergen exposure modifies the relation of sensitization to fraction of exhaled nitric oxide levels in children at risk for allergy and asthma. *J Allergy Clin Immunol* 2011;127:1165-72.e5.
38. Melén E, Standl M, Gehring U, Altug H, Antó JM, Berdel D, et al. Air pollution and IgE sensitization in 4 European birth cohorts—the MeDALL project. *J Allergy Clin Immunol* 2021;147:713-22.
39. Chawes BL, Wolsk HM, Carlsson CJ, Rasmussen MA, Følsgaard N, Stokholm J, et al. Neonatal airway immune profiles and asthma and allergy endpoints in childhood. *Allergy* 2021;76:3713-22. <https://doi.org/10.1111/all.14862>.
40. Lee PC, Talbott EO, Roberts JM, Catov JM, Sharma RK, Ritz B. Particulate air pollution exposure and C-reactive protein during early pregnancy. *Epidemiology* 2011;22:524-31. <https://doi.org/10.1097/EDE.0b013e31821c6c58>. Erratum in: *Epidemiology* 2011;22:752.
41. Roumeliotaki T, Vafeiadi M, Margetaki K, Chalkiadaki G, Dermizaki E, Kogevinas M, et al. Association between prenatal exposure to air pollution and inflammatory markers in children. *Environ Epidemiol* 2019;3:403.
42. Dadvand P, Nieuwenhuijsen MJ, Agustí À, de Batlle J, Benet M, Beelen R, et al. Air pollution and biomarkers of systemic inflammation and tissue repair in COPD patients. *Eur Respir J* 2014;44:603-13.
43. Gao N, Xu W, Ji J, Yang Y, Wang ST, Wang J, et al. Lung function and systemic inflammation associated with short-term air pollution exposure in chronic obstructive pulmonary disease patients in Beijing, China. *Environ Health* 2020;19:12.
44. Hamanaka RB, Mutlu GM. Particulate matter air pollution: effects on the cardiovascular system. *Front Endocrinol* 2018;9:680.
45. Lee BJ, Kim B, Lee K. Air pollution exposure and cardiovascular disease. *Toxicol Res* 2014;30:71-5.
46. Breton CV, Mack WJ, Yao J, Berhane K, Amadeus M, Lurmann F, et al. Prenatal air pollution exposure and early cardiovascular phenotypes in young adults. *PLoS One* 2016;11:e0150825.
47. Schneider D, Hong JY, Bowman ER, Chung Y, Nagarkar DR, McHenry CL, et al. Macrophage/epithelial cell CCL2 contributes to rhinovirus-induced hyperresponsiveness and inflammation in a mouse model of allergic airways disease. *Am J Physiol Lung Cell Mol Physiol* 2013;304:L162-9.
48. Mostafavi N, Jeong A, Vlaanderen J, Imboden M, Vineis P, Jarvis D, et al. The mediating effect of immune markers on the association between ambient air pollution and adult-onset asthma. *Sci Rep* 2019;9:8818.
49. Gomes-Duarte A, Lacerda R, Menezes J, Romão L. eIF3: a factor for human health and disease. *RNA Biol* 2018;15:26-34.
50. Volta V, Pérez-Baos S, de la Parra C, Katsara O, Ernlund A, Dornbaum S, et al. A DAP5/eIF3d alternate mRNA translation mechanism promotes differentiation and immune suppression by human regulatory T cells. *Nat Commun* 2021;12:6979.